Analytical Thinking in Geometry: Patterns & Sequences

Geometry & Mathematical Reasoning Worksheet · Grade 6–9

Name: _____ Date: _____

Learning Objectives

- Use logical thinking to identify and extend numerical patterns in sequences
- Determine the operation (addition, multiplication, etc.) that governs a given sequence
- Predict unknown terms in a sequence by applying discovered rules

Problems

1. Find the next number in the sequence: 2, 4, 6, 8, 10, ___.

2, 4, 6, 8, 10, ?

2. Find the next number in the sequence: 5, 6, 8, 11, 15, ___.

5, 6, 8, 11, 15, ?

3. Find the next number in the sequence: 3, 4, 6, 9, 13, 18, ___.

3, 4, 6, 9, 13, 18, ?

4. Find the next number in the sequence: 2, 6, 18, 54, ___.

2, 6, 18, 54, ?

5. Find the next number in the sequence: 3, 6, 18, 36, 108, 216, ___.

3, 6, 18, 36, 108, 216, ?

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6. A sequence of squares is drawn where the side lengths follow the pattern 1, 3, 5, 7, 9 cm. What is the perimeter of the next square in the pattern?

$$P = 4s,\ s = 11$$

7. The number of triangles in a growing geometric figure follows the sequence 1, 4, 9, 16, 25. What is the next number of triangles, and what pattern rule applies?

$$a_n = n^2$$

8. A sequence of geometric figures has perimeters 8, 12, 16, 20, 24 cm. If the perimeter grows at the same rate, write an expression for the nth perimeter and find the 10th term.

$$P_n = 4n + 4$$

9. A sequence of numbers is given as 96, 48, 24, 12, ___. Identify the operation used and find the next two terms.

$$a_n = 96 \cdot \left(\frac{1}{2}\right)^{n-1}$$

10. Two geometric sequences share the same starting value of 4. Sequence A multiplies by 3 each step, and Sequence B adds 12 each step. After 5 terms, which sequence has a greater 5th term and by how much?

$$A : 4 \cdot 3^{n-1},\ B : 4 + 12(n-1)$$

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Analytical Thinking in Geometry: Patterns & Sequences — Answer Key

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Answer Key

1. Answer: 12
- Observe that each term increases by 2: 4 - 2 = 2, 6 - 4 = 2, etc.
- Add 2 to the last term: 10 + 2 = 12.

2. Answer: 20
- Find the differences: 6-5=1, 8-6=2, 11-8=3, 15-11=4. The differences increase by 1 each time.
- The next difference is 5, so 15 + 5 = 20.

3. Answer: 24
- Differences are: 1, 2, 3, 4, 5 — increasing by 1 each step.
- The next difference is 6, so 18 + 6 = 24.

4. Answer: 162
- Check if multiplication is involved: 6 ÷ 2 = 3, 18 ÷ 6 = 3, 54 ÷ 18 = 3. Each term is multiplied by 3.
- Multiply the last term by 3: 54 × 3 = 162.

5. Answer: 648
- Find the ratios: 6÷3=2, 18÷6=3, 36÷18=2, 108÷36=3, 216÷108=2. The pattern alternates ×2 and ×3.
- Following the pattern, the next operation is ×3: 216 × 3 = 648.

6. Answer: 44 cm
- The side lengths increase by 2 each time: 1, 3, 5, 7, 9 → the next side length is 9 + 2 = 11 cm.
- Perimeter of a square = 4 × side = 4 × 11 = 44 cm.

7. Answer: 36 (the rule is n^2)
- Recognize that $1=1^2$, $4=2^2$, $9=3^2$, $16=4^2$, $25=5^2$ — these are perfect squares.
- The next term is $6^2 = 36$.

8. Answer: 44 cm
- The perimeters increase by 4 each time (arithmetic sequence). First term is 8, common difference is 4.
- Formula: P_n = 8 + (n - 1) × 4 = 4n + 4. For n = 10: P_10 = 4(10) + 4 = 44 cm.

9. Answer: 6 and 3
- Check the ratio: 48÷96 = 1/2, 24÷48 = 1/2, 12÷24 = 1/2. Each term is multiplied by 1/2 (divided by 2).
- Next two terms: 12 ÷ 2 = 6, and 6 ÷ 2 = 3.

10. Answer: Sequence A is greater by 272

• Sequence B: 4, 16, 28, 40, 52. The 5th term is 4 + 12(4) = 52. Difference: 324 - 52 = 272, so Sequence A is greater by 272.

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